

Task 1: Data Cleaning and Preprocessing

Project Overview

This project demonstrates data cleaning and preprocessing techniques on the **Netflix Movies and TV Shows** dataset as part of the Data Analyst Internship at DataX Labs.

Objective: Clean and prepare a raw dataset containing nulls, duplicates, and inconsistent formats to make it analysis-ready.

Tools Used: Python (Pandas)

Dataset Information

Attribute	Details
Source	Netflix Movies and TV Shows (Kaggle)
Original Size	5,967 rows × 13 columns
Final Size	5,967 rows × 13 columns
File Format	CSV

Columns

Column	Description
show_id	Unique identifier for each title
title	Name of the movie/TV show
description	Brief summary of the content
director	Director(s) of the content
genres	Genre categories
cast	Main cast members
production_country	Country where the content was produced
release_date	Year of release
rating	Content rating (TV-MA, TV-14, etc.)
duration	Duration in minutes or seasons
imdb_score	IMDb rating (out of 10)
content_type	Movie or TV Show
date_added	Date when content was added to Netflix

Data Cleaning Steps Performed

1. Column Renaming

- Standardized all column names to **snake_case** (lowercase with underscores)
- Example: Show Id → show_id, Production Country → production_country

2. Handling Missing Values

Column	Missing Count	Treatment
director	2,064	Filled with "Unknown"
cast	530	Filled with "Not Available"
production_country	559	Filled with "Unknown"
release_date	3	Filled with mode (2019)
rating	4	Filled with "Not Rated"
duration	3	Filled with "Unknown"
imdb_score	608	Extracted numeric value, filled with median (6.7)
date_added	1,335	Filled with "Not Available"

Total Missing Values Resolved: 5,106 → 0

3. Data Type Corrections

Column	Before	After
release_date	float64	int64
imdb_score	object (text)	float64

4. Date Format Standardization

- Converted date_added from "Month Day, Year" format to "dd-mm-yyyy"
- Example: "September 8, 2020" → "08-09-2020"

5. Text Standardization

- Stripped leading/trailing whitespace from all text columns
- Ensured consistent formatting across categorical fields

6. Duplicate Check

- Full row duplicates checked: **0 found**
- Duplicate show_id entries checked: **0 found**

7. Outlier Validation

Column	Range	Status
release_date	1925 -2021	✔ Valid
imdb_score	1.3 -9.4	✔ Valid

 Final Data Quality Report

Quality Check	Status
✔ No Missing Values	100% complete
✔ Correct Data Types	All columns properly typed
✔ Standardized Columns	snake_case format
✔ Uniform Date Format	dd-mm-yyyy
✔ Clean Text Values	Whitespace removed
✔ No Duplicates	Dataset is unique
✔ Realistic Value Ranges	Outliers validated

Dataset Status: ✔ **READY FOR ANALYSIS & MODELING**

 Files in This Repository

File	Description
netflixData_cleaned.csv	The cleaned and preprocessed dataset
data_cleaning_summary.txt	Detailed text summary of all changes
README.md	This file - project documentation

 How to Use

```
import pandas as pd

# Load the cleaned dataset
df = pd.read_csv('netflixData_cleaned.csv')

# Check the data
print(df.head())
print(df.info())
```

 Key Takeaways

- 1. **Missing values** are common in real-world datasets and require strategic handling (imputation vs. removal)
- 2. **Data types** must be corrected before analysis to prevent errors
- 3. **Standardization** ensures consistency and improves data quality
- 4. **Date formatting** should be uniform for time-based analysis
- 5. **Duplicate checking** is essential for data integrity

Interview Questions Covered

1.  What are missing values and how do you handle them?
 2.  How do you treat duplicate records?
 3.  Difference between `dropna()` and `fillna()` in Pandas?
 4.  What is outlier treatment and why is it important?
 5.  Explain the process of standardizing data
 6.  How do you handle inconsistent data formats (e.g., date/time)?
 7.  What are common data cleaning challenges?
 8.  How can you check data quality?
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Author

Data Analyst Internship - Task 1 Submission

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This project was completed as part of the Data Analyst Internship program.